

Solutions to Worksheet #3

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1. **Solution.** The semistandard tableaux of shape $(2, 2)$ give

$$s_{2,2} = m_{2,2} + m_{2,1,1} + 2m_{1,1,1,1}.$$

For example, the coefficient 2 is the number of standard Young tableaux of shape $(2, 2)$.

The Jacobi–Trudi identity gives

$$s_{2,2} = h_2^2 - h_3 h_1 = h_{2,2} - h_{3,1}.$$

Since $(2, 2)$ is self-conjugate and the involution ω exchanges h_r with e_r , we also have

$$s_{2,2} = e_2^2 - e_3 e_1 = e_{2,2} - e_{3,1}.$$

Finally, using the character values for the representation indexed by $(2, 2)$, or converting the preceding formulas with Newton's identities, gives

$$s_{2,2} = \frac{1}{12}p_{1,1,1,1} + \frac{1}{4}p_{2,2} - \frac{1}{3}p_{3,1}.$$

2. **Solution.**

- (a) Suppose a semistandard tableau of shape λ and content μ exists. Because columns strictly increase, an entry in row r is at least r . Therefore all entries among $1, 2, \dots, r$ must lie in the first r rows. It follows that

$$\mu_1 + \dots + \mu_r \leq \lambda_1 + \dots + \lambda_r \quad \text{for every } r.$$

If $\mu >_{\text{lex}} \lambda$, let r be the first index at which they differ. Then $\mu_i = \lambda_i$ for $i < r$ and $\mu_r > \lambda_r$, contradicting the displayed inequality. Hence

$$\mu >_{\text{lex}} \lambda \implies K_{\lambda, \mu} = 0.$$

- (b) In a tableau of shape and content λ , all λ_1 copies of 1 must lie in the first row, so that row consists entirely of 1's. After removing it, all λ_2 copies of 2 must fill the second row. Continuing in this way shows that row i must consist entirely of i 's. This tableau is semistandard and is the only possibility. Thus

$$\boxed{K_{\lambda,\lambda} = 1.}$$

3. **Solution.** The monomial expansion of a Schur function is

$$s_\lambda = \sum_{\mu \vdash k} K_{\lambda,\mu} m_\mu.$$

Order the partitions of k lexicographically. Exercise 2 says that this change-of-basis matrix is triangular, and its diagonal entries are all $K_{\lambda,\lambda} = 1$. Its determinant is therefore 1, so it is invertible. Since the monomial functions $\{m_\mu : \mu \vdash k\}$ form a basis of Λ^k , the same is true of

$$\boxed{\{s_\lambda : \lambda \vdash k\}.}$$

4. **Solution.** Only the entries 1 and 2 are available. Since columns strictly increase, a tableau can have at most two rows. Its shape must be

$$(2k - r, r) \quad \text{for some } 0 \leq r \leq k.$$

In the r columns of height two, the top entry must be 1 and the bottom entry must be 2. The remaining cells are all in the first row; weak increase forces the remaining $k - r$ copies of 1 to come before the remaining $k - r$ copies of 2. Thus there is exactly one tableau for each value of r . The answer is

$$\boxed{k + 1.}$$

5. **Solution.** Let

$$E(t) = \sum_{n \geq 0} e_n t^n, \quad H(t) = \sum_{n \geq 0} h_n t^n.$$

The given identity is exactly the statement

$$E(-t)H(t) = 1.$$

Suppose ω is defined by $\omega(e_n) = h_n$ and extended as an algebra homomorphism. Apply ω to the displayed identity:

$$H(-t)\omega(H(t)) = 1.$$

Replacing t by $-t$ in the original identity also gives $E(t)H(-t) = 1$. An inverse is unique, so

$$\omega(H(t)) = E(t).$$

Comparing coefficients yields $\omega(h_n) = e_n$. Thus ω^2 fixes every e_n (and every h_n). Since the e_n generate the ring of symmetric functions,

$$\boxed{\omega^2 = \text{id.}}$$

6. **Solution.** Since $h_r = s_r$, apply the Pieri rule to $s_3 h_2$. Adding a horizontal strip of two boxes to the row (3) produces exactly (5), (4, 1), and (3, 2). Hence

$$\boxed{h_{3,2} = s_5 + s_{4,1} + s_{3,2}.$$

7. **Solution.** Apply ω to the answer in Exercise 6. The involution exchanges h and e and sends s_λ to $s_{\lambda'}$. The conjugates of (5), (4, 1), and (3, 2) are respectively (1^5) , $(2, 1^3)$, and $(2, 2, 1)$. Therefore

$$\boxed{e_{3,2} = s_{1^5} + s_{2,1^3} + s_{2,2,1}.$$

8. **Solution.** Newton's identities give

$$p_3 = 3h_3 - 3h_2 h_1 + h_1^3.$$

By the Pieri rule,

$$h_3 = s_3, \quad h_2 h_1 = s_3 + s_{2,1}, \quad h_1^3 = s_3 + 2s_{2,1} + s_{1,1,1}.$$

Substitution gives

$$\boxed{p_3 = s_3 - s_{2,1} + s_{1,1,1}.$$

9. **Solution.** The statement is false. Let $\lambda = (2)$ and $\mu = (1, 1)$. There is one semistandard tableau of shape (2) and content (1, 1), namely a row containing 1, 2. Thus $K_{(2),(1,1)} = 1$. A column of shape (1, 1) cannot be filled with two copies of 1 because columns must strictly increase, so $K_{(1,1),(2)} = 0$. Hence the two Kostka numbers need not be equal.
10. **Solution.** The partitions of 3 give

$$\boxed{\begin{aligned} h_3 &= s_3, \\ h_{2,1} &= s_3 + s_{2,1}, \\ h_{1,1,1} &= s_3 + 2s_{2,1} + s_{1,1,1}. \end{aligned}}$$

The pattern is Young's rule:

$$\boxed{h_\lambda = \sum_{\mu \vdash 3} K_{\mu, \lambda} s_\mu.$$

In general, the coefficient of s_μ in h_λ is the number of semistandard tableaux of shape μ and content λ .

11. **Solution.** For $a \geq 1$ and $b \geq 0$, the Jacobi–Trudi determinant for the hook $(a, 1^b)$ is

$$s_{a,1^b} = \begin{vmatrix} h_a & h_{a+1} & h_{a+2} & \cdots & h_{a+b} \\ 1 & h_1 & h_2 & \cdots & h_b \\ 0 & 1 & h_1 & \cdots & h_{b-1} \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & 1 & h_1 \end{vmatrix}.$$

Expand along the first row. The minor corresponding to h_{a+i} is the Jacobi–Trudi determinant for e_{b-i} ; the cofactor sign is $(-1)^i$. Therefore

$$s_{a,1^b} = \sum_{i=0}^b (-1)^i h_{a+i} e_{b-i}.$$

This also proves the formula inductively: the case $b = 0$ is $s_a = h_a$, and increasing b adds one more cofactor to the same Jacobi–Trudi expansion.